

PRASANNA KUMAR GANESAN

Lead Data Engineer | Chennai, India

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PROFESSIONAL SUMMARY

Lead Data Engineer with 9+ years of experience in designing scalable data engineering solutions, ETL/ELT pipelines, and Azure lakehouse platforms using Azure Databricks, Azure Data Factory (ADF), PySpark, and Delta Lake. Expertise in data modelling, Spark performance tuning, cloud migrations, and production CI/CD for enterprise data platforms across banking, fintech and manufacturing domains. Experienced in implementing modern Medallion (Bronze–Silver–Gold) architectures and governed data platforms using Unity Catalog.

TECHNICAL SKILLS

Cloud Platforms: Azure Databricks, Azure Data Factory (ADF), Azure Data Lake Storage Gen2 (ADLS Gen2), Azure Devops

Data Engineering: ETL, ELT, Lakehouse Architecture, Delta Lake, Delta Live Tables (DLT), Medallion Architecture, Data Modelling, Data Warehousing

Programming: PySpark, Python, SQL, Spark SQL, Unix, PL/SQL, PL/pgSQL

Databases: Oracle, PostgreSQL

Orchestration & CI/CD: Lakeflow Declarative Pipelines, Lakeflow Jobs, Databricks Asset Bundles (DAB), Git, GitHub Actions, GitLab CI/CD

Governance & Optimization: Unity Catalog, Query Optimization, Performance Tuning, Partitioning, Z-Ordering, Vacuum, Liquid Clustering

PROFESSIONAL EXPERIENCE

Senior Associate – Cognizant | Apr 2021 – Present

- **Architected and delivered ETL/ELT platforms** — Azure Data Factory, PySpark, ADLS Gen2; automated production data flows and improved analytics availability.
- **Led architecture and rollout** of a Bronze–Silver–Gold Lakehouse on **Azure Databricks** with **Delta Lake**, standardizing ingestion and supporting high-volume reporting.
- **Orchestrated Spark performance improvements** with partitioning, caching, Z-ordering, clustering, and query tuning, reducing pipeline runtime **50%** and lowering compute costs.
- **Established data governance** with **Unity Catalog**, enabling end-to-end lineage and RBAC to secure sensitive datasets and support audits.
- Developed AI-based document processing solution using Databricks IDP (AI_EXTRACT, AI_PARSE) for unstructured PDF ingestion, achieving ~60% reduction in manual processing effort, ~80% error reduction, and ~120 hours/month operational savings in pilot deployment.

- **Deployed production ETL orchestration and CI/CD** — Lakeflow Declarative Pipelines, Lakeflow Jobs, DLT, and Databricks Asset Bundles (DAB) for versioned deployments.

Associate Consultant – Capgemini | Jan 2019 – Mar 2021

- **Automated data ingestion** via SOAP APIs/web services, reducing manual effort by **80%**.
- **Migrated Oracle Forms 10g → 12c**, achieving **99.9% uptime** during modernization.
- **Optimized queries and indexing** using execution-plan analysis to improve query performance.
- **Migrated databases on-prem → OCI**, reducing infrastructure costs **30%**.
- **Designed and maintained relational data models** to ensure data integrity and normalization.
- **Developed and supported data integration workflows** for reliable validation, transformation, and delivery across enterprise systems.

Senior Process Associate – TCS | Aug 2016 – Jan 2019

- **Performed technical design and analysis**, producing schema designs, data mappings, and implementation specs for development and QA.
- **Optimized SQL and indexing**, reducing query execution time **40%** through execution-plan analysis and targeted tuning.
- **Migrated Oracle → PostgreSQL** (schema conversion, data migration scripts, validation, and cutover support) as a core developer on the migration team.
- **Developed stored procedures and functions** in **PL/SQL / PL/pgSQL** to encapsulate business logic and improve maintainability.

ACHIEVEMENTS

- Cummins Ideathon Award (Top 2 among 500)
- Impact Award for ETL optimization
- Star Award for performance

EDUCATION

B.E. – Electronics & Communication Engineering
Sri Ramakrishna Institute of Technology (2010-2014)